

Pongporn (Ken) Supa

✉ pongporn.supa@gmail.com ☎ 0788 725 40 75 🌐 Personal website in pongporn-supa 🐙 Github

Robotics engineer with 2.5+ years of experience integrating and deploying the full aerial autonomy stack. Proven expertise in developing reliable middleware and interfaces (ROS2, MAVLink, MAVROS, uXRCE-DDS) for PX4-based UAVs, deploying to embedded hardware and tuning control systems

Education

University of Bristol

MSc in Robotics (74/100, Distinction)

Bristol, UK

Sept 2024 – Sept 2025

- **Focus:** Aerial Navigation and Soft robotics | Advisor: [Dr. Basaran Bahadir Kocer](#) 📧

Chulalongkorn University

BEng in Mechanical Engineering (3.63/4.00, First Class Honour)

Bangkok, TH

Aug 2017 – Aug 2021

- **Focus:** Mechatronics and Control System

Work Experience

Research Assistant: *Bumper Drone: Elastic Morphology Design for Aerial*

Physical Interaction under [Dr. Basaran Bahadir Kocer](#) 📧

Bristol, UK

Sept 2025 – Current

School of Engineering Mathematics and Technology

- Designed elastic collision-resilient drone horns that enabled repeatable passive bumping in GPS-denied and visually degraded environments, improving pitch-attitude oscillation stability by 38 percent
- Developed and tuned PX4 attitude and position control algorithms, integrating of optical flow and downward-facing LiDAR sensors
- Implemented real-time digital signal processing pipelines in ROS 2 and developed closed-loop control using tactile feedback via MAVROS

Robotics Software Engineer

AI and Robotics Ventures, HORRUS project

Bangkok, TH

Apr 2022 – Aug 2024

- Engineered a real-time MAVLink-based middleware, implementing flows for heartbeats, parameters, mission management, precision landing to coordinate docking stations and boosted autonomous mission reliability to 95 percent
- Implemented and designed end-to-end framework for coverage path planning for aerial photogrammetry with geofence constraints using Hierarchical Coverage Path Planning based on sensor footprint
- Built high-throughput backend APIs using Flask and FastAPI for multi-drone video streaming, telemetry monitoring, and real-time teleoperation
- Integrated YOLOv8 vehicle detection and tracking models into an automated alert pipeline

Research Assistant: *Modular Temperature-Controlled Syringe Pump Extruder for Soft Material 3D Printer* under [Assoc. Prof. Ratchatin Chancharoen](#) 📧

Bangkok, TH

May 2021 – Oct 2021

Department of Mechanical Engineering, Chulalongkorn University

- Transformed a cartesian robot into the syringe-based soft material 3D printer, which is capable of controlling the temperature of the syringe and printing bed, helping secure £67K funding from the National Research Council of Thailand (NRCT)
- Designed PCBs for soft material printing benefits, including a high-temperature extruder and dual-temperature controller printing bed to address complex behavior of viscoelastic materials
- Extended open-source 3D printer firmware for adaptive printing based on real-time shape detection, showcased at Thailand Inventor's Day

Publications

- [Submitted to RoboSoft's 26]: P. Supa, A. J. Dunnett, F. Xiao, R. Wu, M. Kovac, B. B. Kocer, "Bumper Drone: Elastic Morphology Design for Aerial Physical Interaction", under review

Selected Projects

Multi-Agent Coordination: Adaptive Point-Tracking Control for Heterogeneous Robots

[Github repo](#) [🔗](#)

- Developed cooperative adaptive control algorithms for heterogeneous mobile robots
- Implemented state estimation for mobile robot using IMU with Kalman Filter

Learning from Demonstration on Robot Manipulation

Mar 2025 - Apr 2025

- Built imitation learning models for trajectory generalization and leader-follower teleoperation control system

Search & Rescue in Mobile Robot

[Github repo](#) [🔗](#) [Youtube](#) [🔗](#)

- Designed and developed Finite-State Machine for navigation and pose correction using map priors knowledge

Apple Counting in Orchards

Dec 2024 - Jan 2025

- Designed and implemented an image processing pipeline and YOLO model using OpenCV and PyTorch, leveraging color, geometric, and texture features for accurate apple detection under varying illumination conditions

Teaching experiences

Teaching Assistant at University of Bristol

Jan 2025 - Current

- Demonstrated and led tutorial sessions on data analytics, software development, programming, and algorithms courses

Teacher Assistant in Manufacturing Process Course

Mar 2022

- Prepared courseware/engineering workshops. Lectured on how 3D printing settings and parameters affect printing quality

Skills

Programming languages: Python, C++, MATLAB, Simulink, Javascript

Software: ROS2, Git, MAVLink, MAVSDK, MAVROS, uXRCE-DDS, OpenCV, PyTorch, PX4, QGroundControl, RViz, Gazebo, Docker, AWS EC2/ECR/S3

Hardware: Proficient in digital fabrication for rapid prototyping, UAV prototyping, PCB design, CAD/Fusion360, Embedded systems/NuttX/STM32/ESP32, Embedded Linux/Jetson nano/Raspberry Pi

Honors & Awards

Bristol Award Plus, University of Bristol

Mar 2025

Think Big Scholarship Award Recipient, University of Bristol

Sept 2024

PTTEP Innovation Award, Silver Award, Autonomous Offshore Drone System

July 2024

Spotlight on New Technology Awards, OTC Asia 2024, AI and Robotics Solutions for Offshore Digitalisation

Feb 2024

Invited speaker on Robotics & AI Bootcamp, Chula Engineering NITAD 18

Mar 2022